Generative City-Wallet: Mobile & AR Spatial Experience

📖 Overview & Vision

This project is a next-generation city wallet that connects people to the most relevant local offers in real-time. Instead of showing static coupons, it acts as a "living city interface"—understanding the moment, predicting likely intent, and generating situation-aware offers that feel useful right now rather than generic or delayed.

The experience empowers local merchants to compete with the dynamic personalization normally exclusive to large digital platforms. Crucially, it achieves this while respecting user privacy by keeping sensitive personal interpretation on-device via Small Language Models (SLMs), only sharing abstract intent signals with the backend.

⚠️ The Problem

Today, many local businesses have access to useful signals, but they lack the personalization infrastructure to act on them in real time. Traditional coupon systems are static, broad, and often irrelevant. There is a fundamental gap between a person's immediate context (e.g., a cold 12-minute lunch break) and a perfectly relevant local offer standing just 80 meters away.

📱 Proposed Experience: Two Surfaces

This product is a mobile-first AI wallet with an AR companion layer.

- **1. Mobile Interface (The Deep Interaction Surface):** The primary surface for the full experience. It handles notification entry points, in-app offer cards, redemption, wallet history, cashback, consent controls, and saved preferences. Designed for fast understanding.
- **2. AR Glasses Interface (The Ambient Interaction Surface):** A lightweight companion that turns the city into the interaction layer. It surfaces subtle visual cues (e.g., "quiet café 80m ahead," or "offer expires in 6 minutes") as glancing prompts to help the user move from awareness to action without opening a dense interface.

🏗️ Three Major Product Modules

A. Context Sensing Layer

This layer brings together live signals to define the composite context state. Inputs are configurable rather than hardcoded.

- Must incorporate at least two real context signal categories visible to the user.
- Aggregates weather data, time-of-day, and local event context.
- Tracks geo-fenced user location.
- Monitors nearby merchant demand or transaction-density proxies (simulated Payone data).

B. Generative Offer Engine

The intelligence layer that turns context into an actual offer. The offer must be generated dynamically, not retrieved from a static database.

- Decides whether a moment is worth acting on.
- Generates the message, tone, and framing of the value proposition dynamically.
- Selects the right amount of urgency.
- Produces the visual card or widget dynamically using Generative UI (GenUI).

C. Seamless Checkout and Redemption

Closes the loop between recommendation and value.

- Manages offer acceptance or dismissal.
- Handles dynamic token or QR generation.
- Executes a seamless simulated checkout via QR scan, token validation, or a cashback mechanic.
- Logs confirmations and performance metrics for the merchant.

🧠 The Privacy Wall: Role of SLMs

Small Language Models are the core of our GDPR-friendly product behavior. **Rule: Raw personal behavior stays local; abstract intent travels.**

- Interpret private user context locally on the device.
- Summarize likely intent (e.g., "browsing," "cold and wants warm stop") without exposing raw sensitive data.
- Personalize the tone and framing of the offer entirely on the edge.

🛣️ Core User Journey

1. **Sensing:** The system senses live context (weather, time, location, merchant demand).
2. **Interpretation:** The on-device SLM interprets what kind of moment this is privately.
3. **Generation:** The generative engine creates a custom offer (copy, design, urgency, incentive).

4. **Delivery:** The offer appears through a deliberate interaction surface (push, in-app card, lock-screen widget, or AR prompt).
5. **Decision:** The user accepts, dismisses, or ignores the offer.
6. **Redemption:** If accepted, the user redeems it through a simulated checkout, QR, token, or cashback flow.
7. **Analytics:** The merchant sees aggregate results in their dashboard.

📊 Merchant Dashboard Scope

The dashboard is a required product module, not an afterthought. It provides privacy-safe insights rather than raw personal tracking.

- **Rule Configuration:** Merchants set simple goals (e.g., "max 20% discount to fill quiet hours") while AI handles execution.
- **Performance Overview:** Displays aggregate offer performance.
- **Metrics:** Accept vs. decline rates, and total redemption counts.
- **Context Triggers:** Explains *why* an offer was generated (e.g., quiet-hour detection + rain).

🎨 UX Principles

Design is the mechanism of acceptance or rejection.

- **The 3-Second Rule:** The offer must be understood instantly, without scrolling or deliberation. This dictates minimal text and strong visual hierarchy.
- **Intentional Channels:** The interaction surface (push, widget, AR) is chosen deliberately based on attention rules.
- **Adaptive Tone:** Messaging shifts dynamically between factual-informative or emotional-situational.
- **Clear Exits:** Expiry, acceptance, or dismissal should feel intentional and leave the user experience intact.

🌟 Example Scenario

A user is walking through the city on a cold Tuesday afternoon with a short lunch break available. Concurrently, a nearby café is experiencing an unusually low transaction volume.

- **Mobile:** The user receives a concise, emotional-situational offer card ("Cold outside? Your cappuccino is waiting.").
- **AR:** A subtle overlay points toward the venue with a short value statement and a 15-minute expiry timer.

🎯 Success Criteria

This project is successful if it demonstrates:

- Real context signals in action responding to a concrete scenario.
- A dynamically generated offer tied to a believable moment.
- A UI designed for 3-second comprehension.
- A complete, connected loop from context detection to simulated checkout.
- An explicit, honest approach to privacy using on-device inference.
- Full payment network integration.
- Large-scale production merchant onboarding.
- Complex, multi-tiered loyalty ecosystems.
- Perfect, production-ready AR hardware integration.
- Advanced personalization based on months of historical user data.